

EC
012

Consumer and producer surplus

Chapter 4

The logo consists of the letters 'EC' stacked above the numbers '012', all in a light blue color. The text is set against a solid dark blue square background.

EC
012

Admin admin

Deadlines:

- Thurs: **LC4**
- Next Tues: **HW4**

Mini-quiz!

Review session w/ TJ

Old Mill 236

Today 1:30–2:30

Friday 1–2

Office hours

Thursday 11:30–
1:30

Old Mill 233

EC 012

Week 2 Agenda

Tuesday

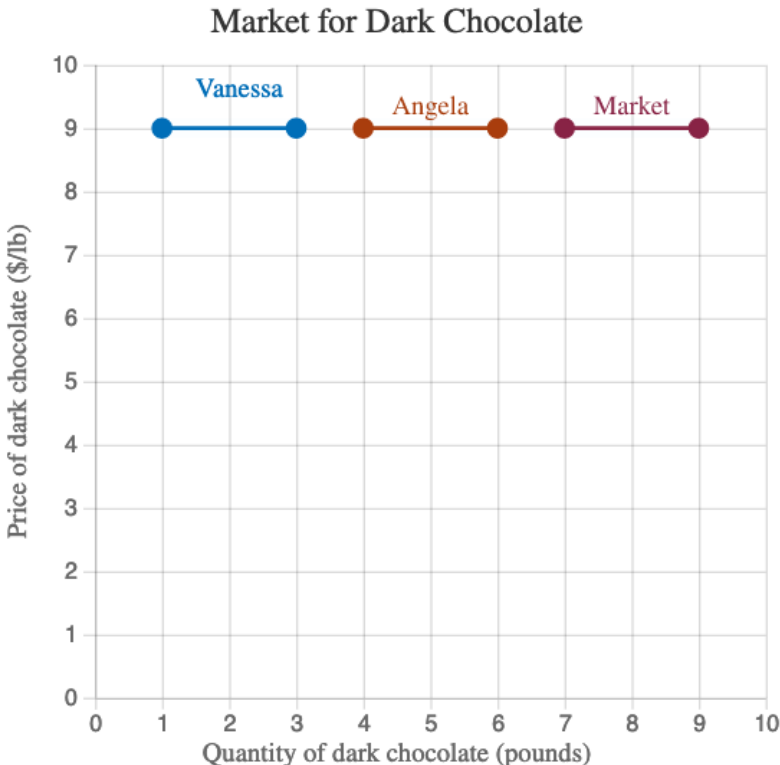
- Review some tricky HW3 questions
- Producer and consumer surplus

Thursday

- Mini-quiz
- More PS/CS

The accompanying table contains the individual demand schedules of dark chocolate for Vanessa and Angela. They are the only people in the market for dark chocolate. Place each of their individual demand curves and the market demand curve at the correct locations on the accompanying graph.

Price of dark chocolate (per pound)	Pounds of dark chocolate demanded by Vanessa	Pounds of dark chocolate demanded by Angela
\$8	1	3
\$1	3	5



After a while, Angela gets sick of eating the bitter chocolate and decides to cut back on dark chocolate and enjoy white chocolate more. What happens as a result of Angela's decision? Select all of the correct answers.

- ☐ The market demand for dark chocolate will decrease.
- ☐ Angela's individual demand for white chocolate increases.
- ☐ Angela's individual demand for dark chocolate decreases.
- ☐ Vanessa's individual demand for dark chocolate increases to compensate for Angela's decreased demand.

During the Obama administration, the development of low-cost batteries for electric cars received large amounts of federal funding in terms of subsidies. Meanwhile, American households gave a higher priority towards minimizing their environmental impact. Consider the market for zero-emissions electric vehicles where there is an upward-sloping supply curve and a downward-sloping demand curve.

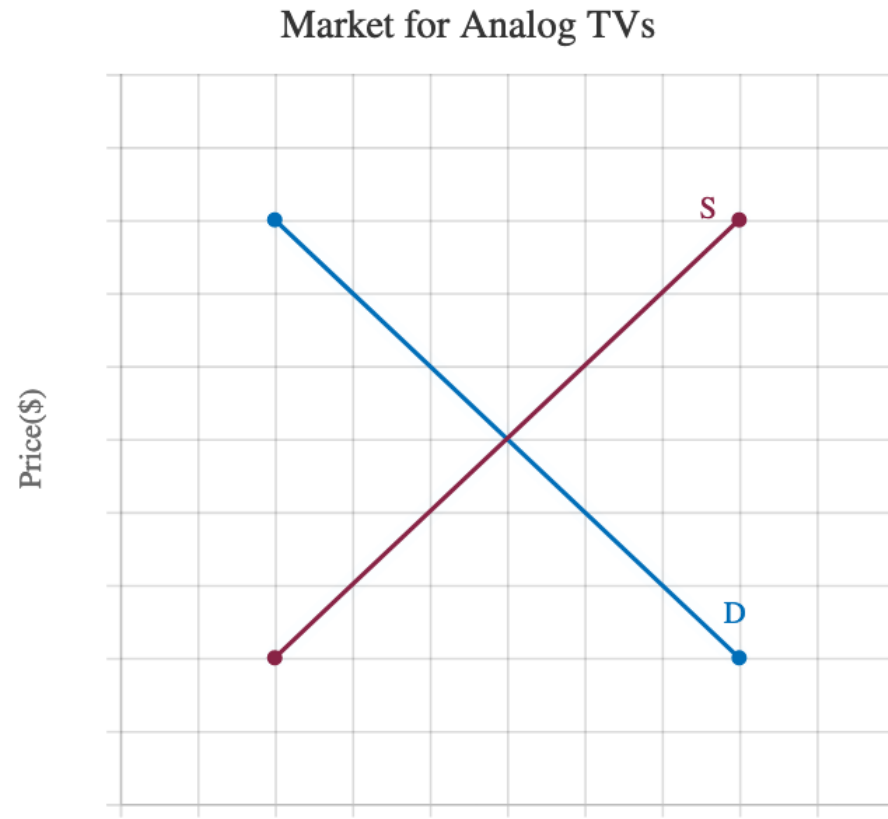
- Which direction will demand and supply shift?
- What will happen to equilibrium quantity?
- What will happen to equilibrium price?

The accompanying graph depicts a hypothetical market for analog TVs.

In June 2009, the U.S. Federal Communications Commission (FCC) mandated that all over-the-air television broadcasts originating in the U.S. must be broadcast digitally instead of the traditional analog signal. These digital transmissions cannot be decoded by traditional analog TVs but require either a digital decoder be attached to an analog TV or a natively digital TV.

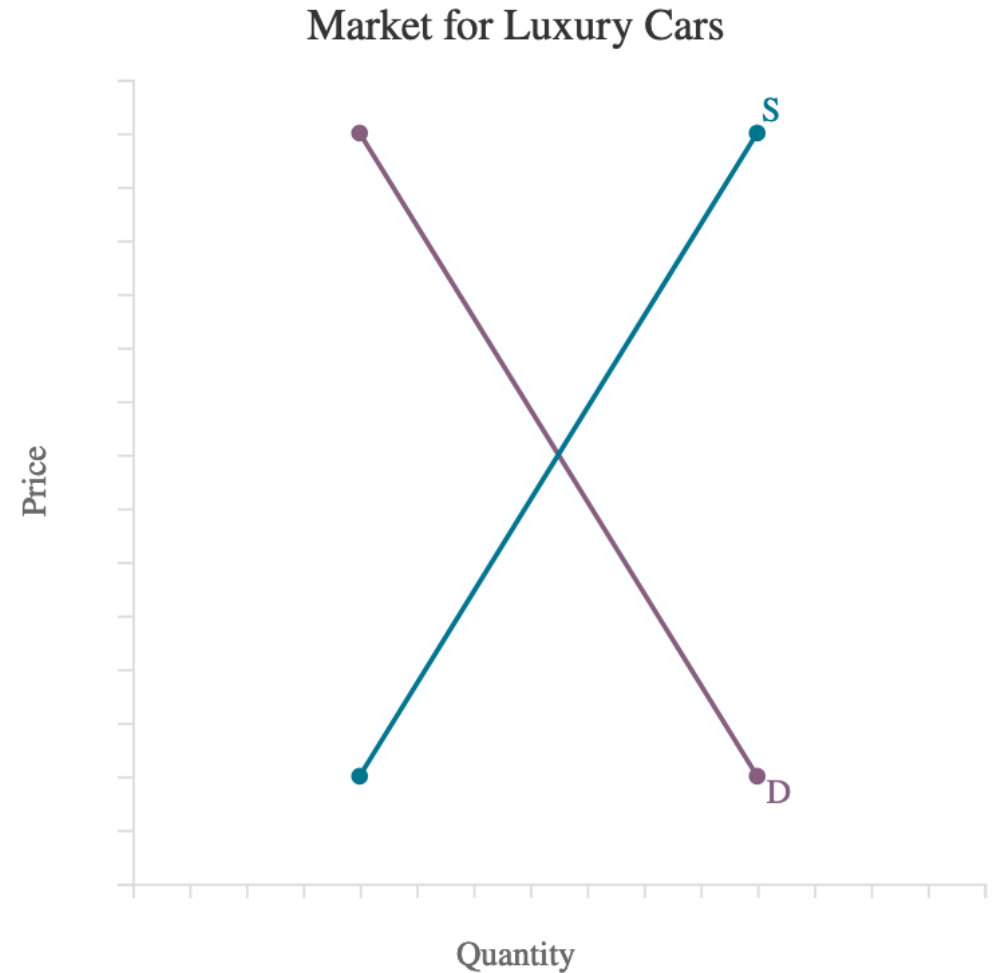
Both sellers and consumers were made aware of the upcoming switch from analog transmission to digital several years before the mandate took effect.

Demonstrate on the graph how the market for analog TVs changed *immediately* following the announcement of the switch to digital. For the supply side, focus on the end seller (such as an electronic store) of a TV and not the manufacturer.



The accompanying graph represents a hypothetical market for luxury automobiles.

Suppose that a major luxury car producer exits the market in order to produce more economy cars. At the same time, a new tax law means that many tax payers will receive large refunds this year. According to a marketing firm, the result is a higher quantity of luxury cars selling at higher prices. Manipulate the graph to demonstrate what must be happening to supply and demand.



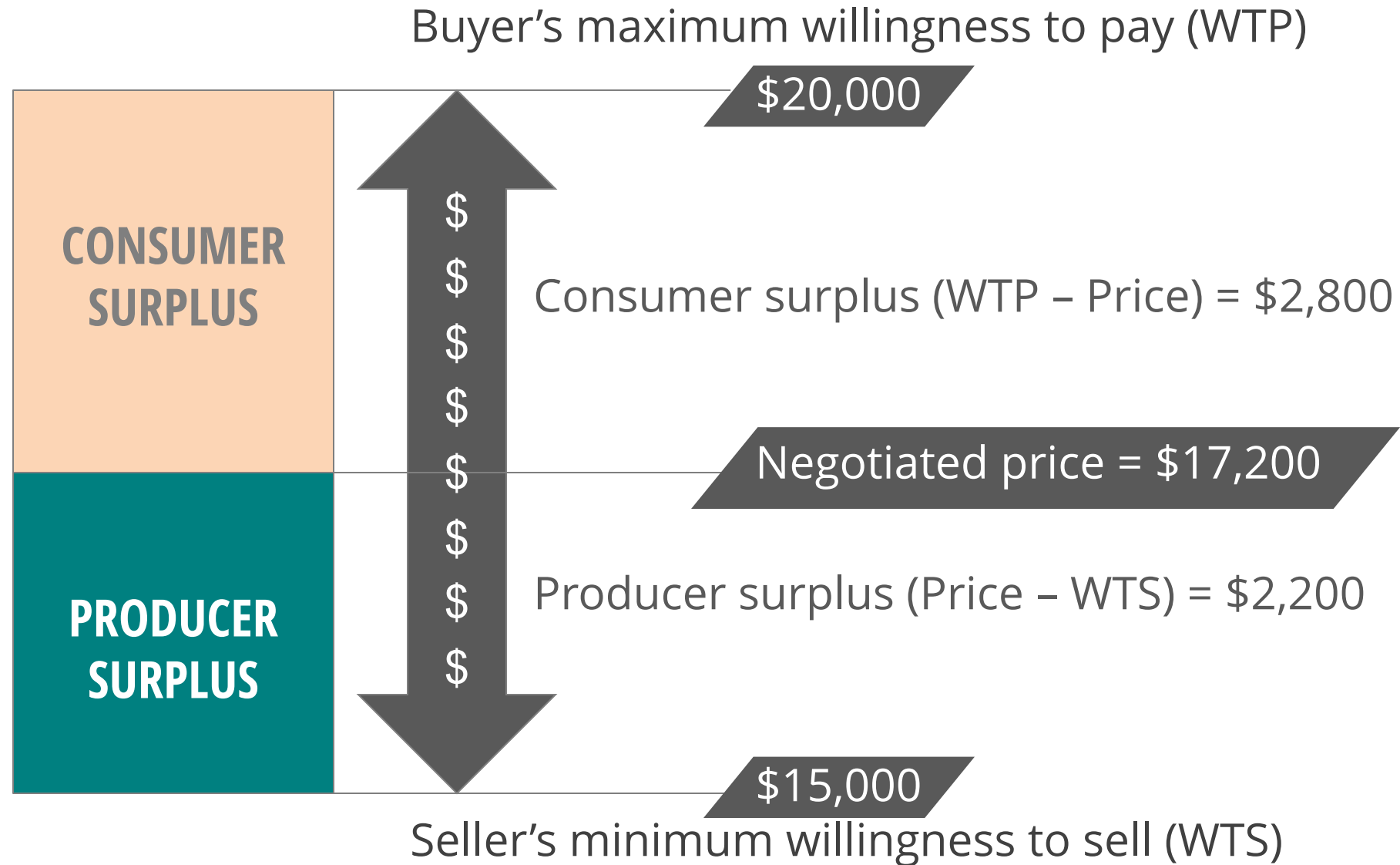
Where are we?

- We just covered **eleven principles of economics (CH1)** semi - **asynchronously**
- We learned how to set up **supply and demand curves** and **move them** around **(CH3)**.

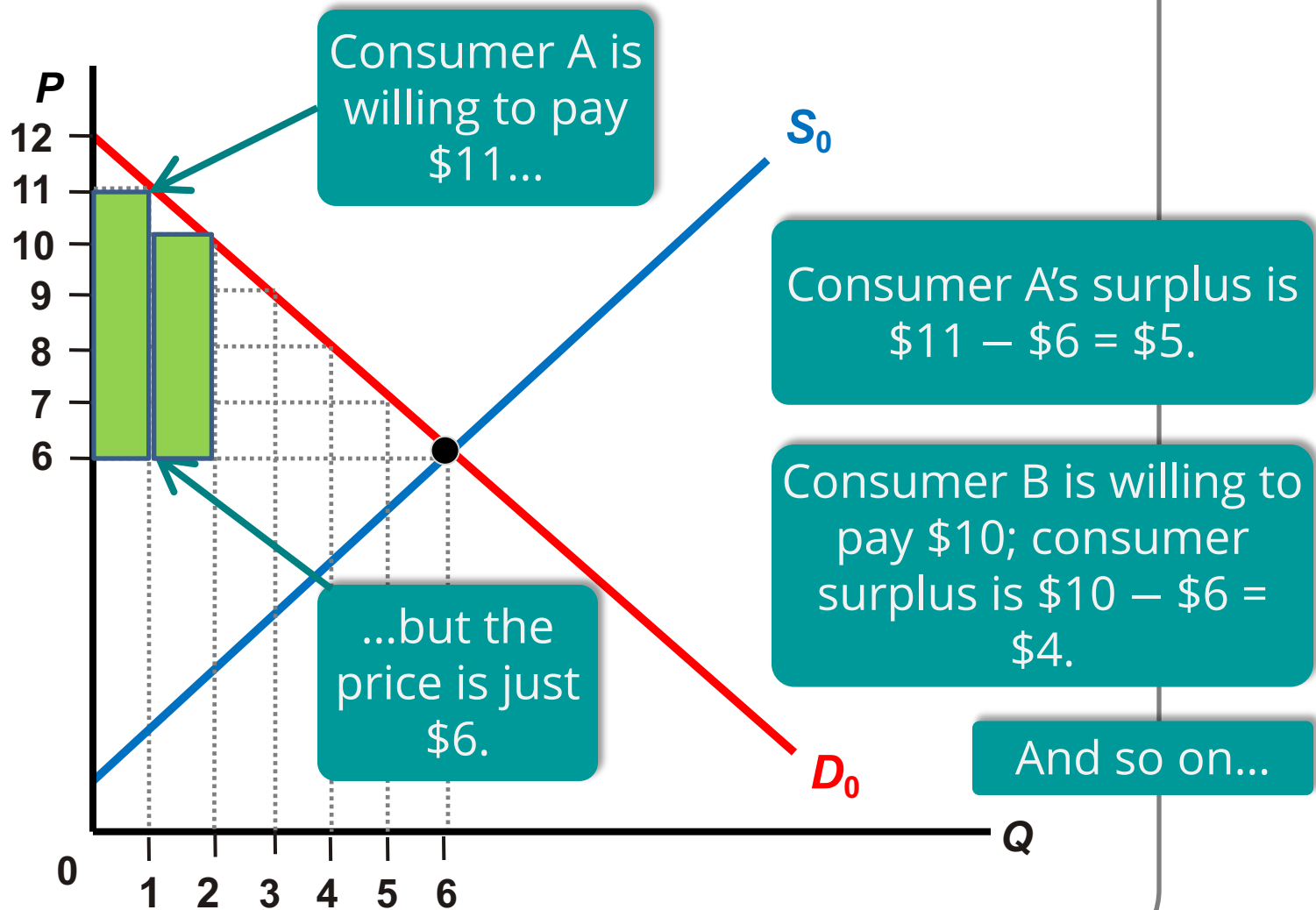
Learning objectives

- Calculate **consumer** and **producer surplus**
- Define and identify **deadweight loss**
- When markets lead to **efficiency**
- Types of **market failures**

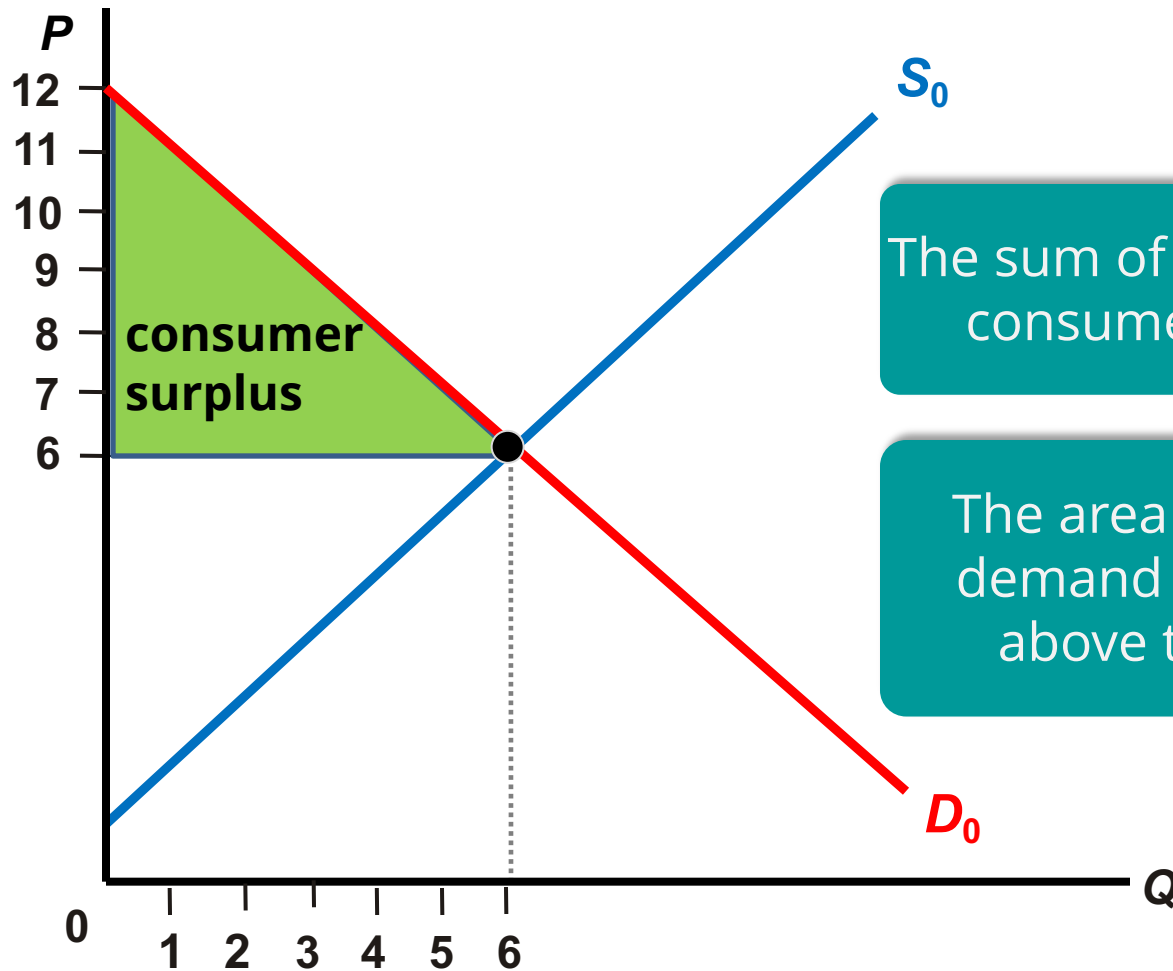
CONSUMER & PRODUCER SURPLUS



INDIVIDUAL CONSUMER SURPLUS



TOTAL CONSUMER SURPLUS



The sum of all individual consumer surplus

The area below the demand curve and above the price

Buy some new shirts, George!

George wants to buy some new shirts for work. He is willing to pay \$35 for the first shirt, \$25 for the second shirt, and \$15 for the third.

Oxford & Co. (his favorite shirt shop!) sells shirts for \$28 each.

What is the efficient number of shirts for George to buy?

- a) 0
- b) 1
- c) 2
- d) 3



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George wants to buy some new shirts for work. He is willing to pay \$35 for the first shirt, \$25 for the second shirt, and \$15 for the third.

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What is George's consumer surplus?

- a) \$0
- b) \$7
- c) \$10
- d) \$28
- e) \$35



Calculating consumer surplus

Mark and Rasheed are buying new calculators for Math 19. Mark is willing to pay \$75 and Rasheed is willing to pay \$100 for a graphing calculator. The price for a calculator at the bookstore is \$65. How much is their total consumer surplus?

- a) \$10
- b) \$35
- c) \$45
- d) \$60

Figure: The Market for Hamburgers



If the price of hamburgers is \$2, consumer surplus will equal:

- a) \$650
- b) \$400
- c) \$225
- d) \$450

If the price of hamburgers is \$2.50 today, what will happen to the price in the coming days/months?

- a) It will fall
- b) It will rise
- c) It will remain the same
- d) Not enough information

If the price of hamburgers rises, what will be the main effect on the chart?

Figure: The Market for Hamburgers



- a) Supply will shift left
- b) Supply will shift right
- c) Demand will shift left
- d) Demand will shift right

If the price of a hamburger is \$2, consumer surplus will be ...

Figure: The Market for Hamburgers



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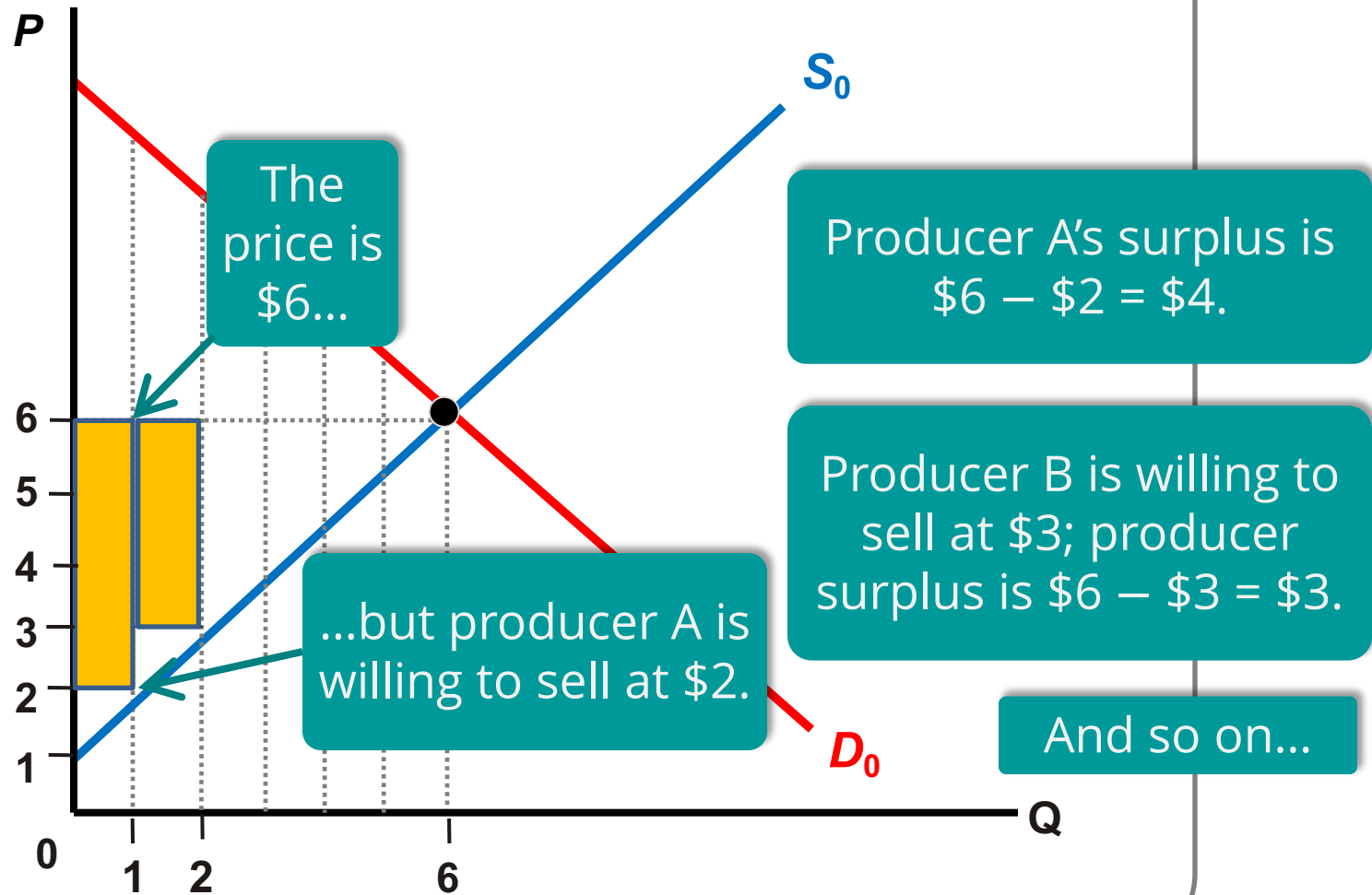
The sky is dumping freezing rain, it's 11:30 at night, and you're standing on a street corner 15 miles from home. You open your Uber app and see that the trip is going to cost you roughly \$50, as prices are surging 200 percent above normal. But you summon the car anyways. In fact, you might not have hesitated to go as high as \$100, just to get dry in that comprising moment.

[Laura Bliss](#), Bloomberg, 15 September 2016

What is your consumer surplus?

What market inefficiency does Uber solve?

INDIVIDUAL PRODUCER SURPLUS



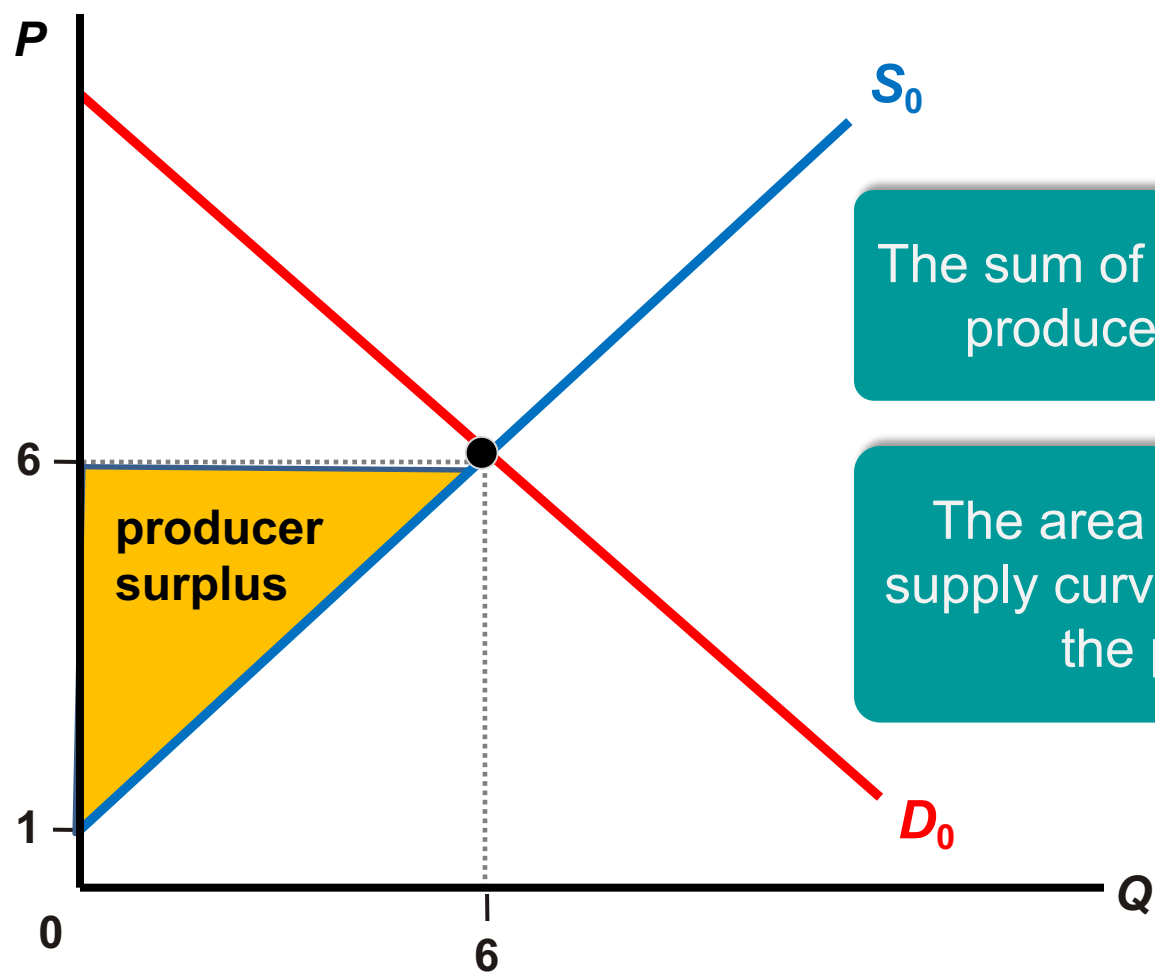
Four different students each have one textbook they might sell back to the bookstore. Below is each student's reservation price: the lowest price at which they would be willing to sell the book. The bookstore is buying textbooks at a price of \$25.

Seller	Seller's Reservation Price
Alex	\$12
Piper	\$16
Daya	\$22
Red	\$30

How many students will sell their textbooks?

- a) 0
- b) 1
- c) 2
- d) 3
- e) 4

TOTAL PRODUCER SURPLUS

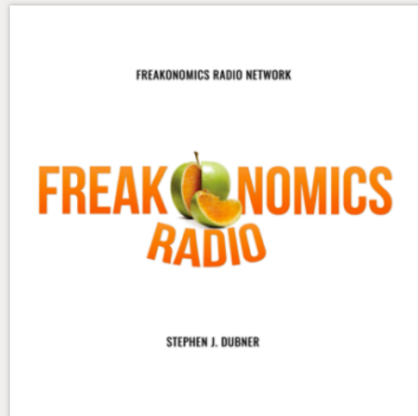


The sum of all individual producer surplus

The area above the supply curve and below the price

**How does Uber's ability to tailor prices affect
producer surplus?**

Why might Uber be “an economist’s dream”?



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EPISODE 258

Why Uber Is an Economist’s Dream

To you, it’s just a ride-sharing app that gets you where you’re going. But to an economist, Uber is a massive repository of moment-by-moment data that is helping answer some of the field’s most elusive questions.



Sep 7, 2016

By **Stephen J. Dubner**
Produced by **Harry Huggins**

 **Comments**

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Figure: The Market for Hamburgers



If 400 hamburgers are sold, producer surplus will equal

- a) \$650
- b) \$400
- c) \$510
- d) \$200

Figure: The Market for Hamburgers



If the price falls from \$1.50 to \$1, what is the loss in producer surplus?

- a) \$50
- b) \$45
- c) \$150
- d) \$200

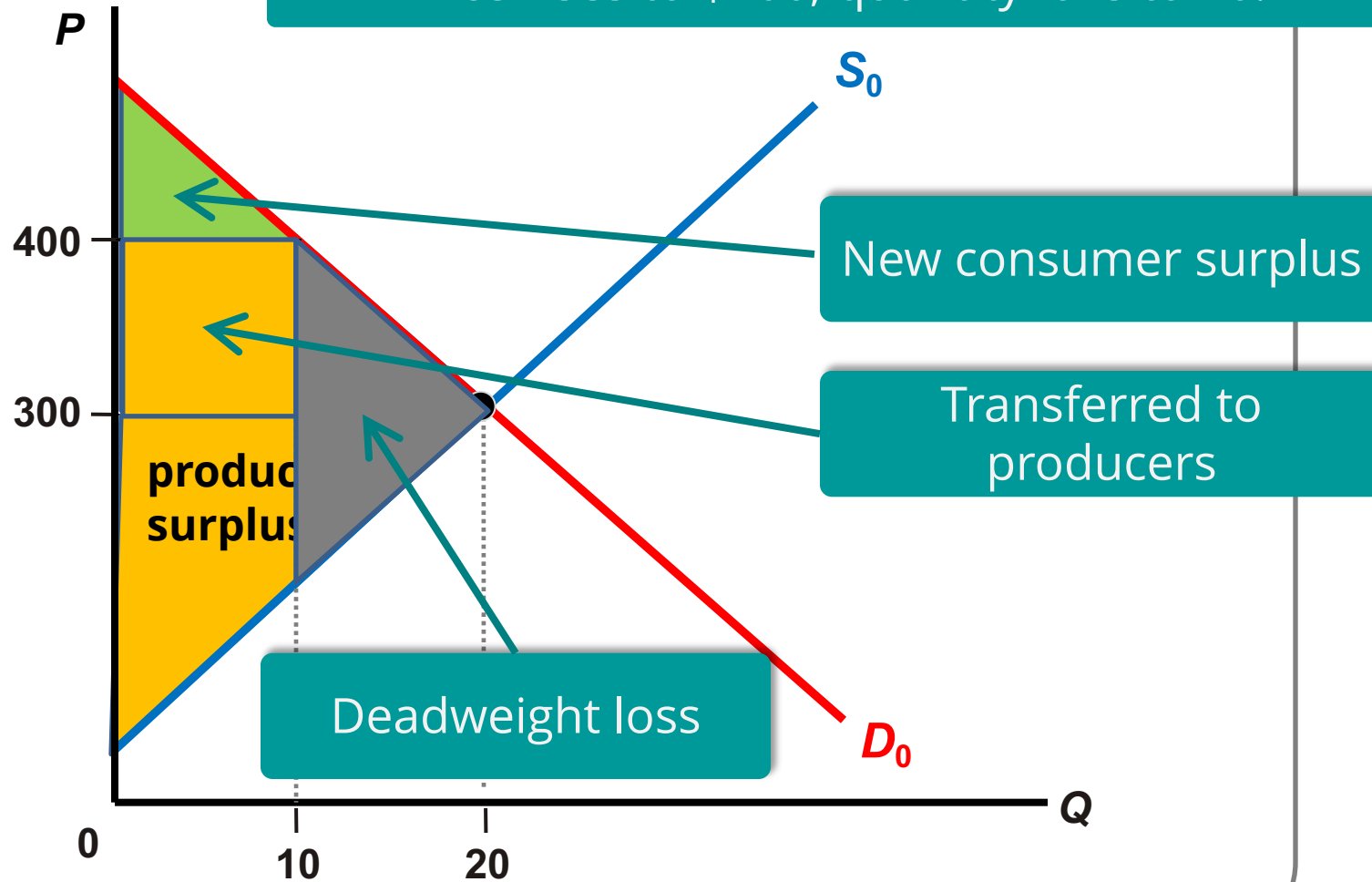
TOTAL SURPLUS

When markets are efficient,
TS is MAXIMIZED!

$$TS = CS + PS$$

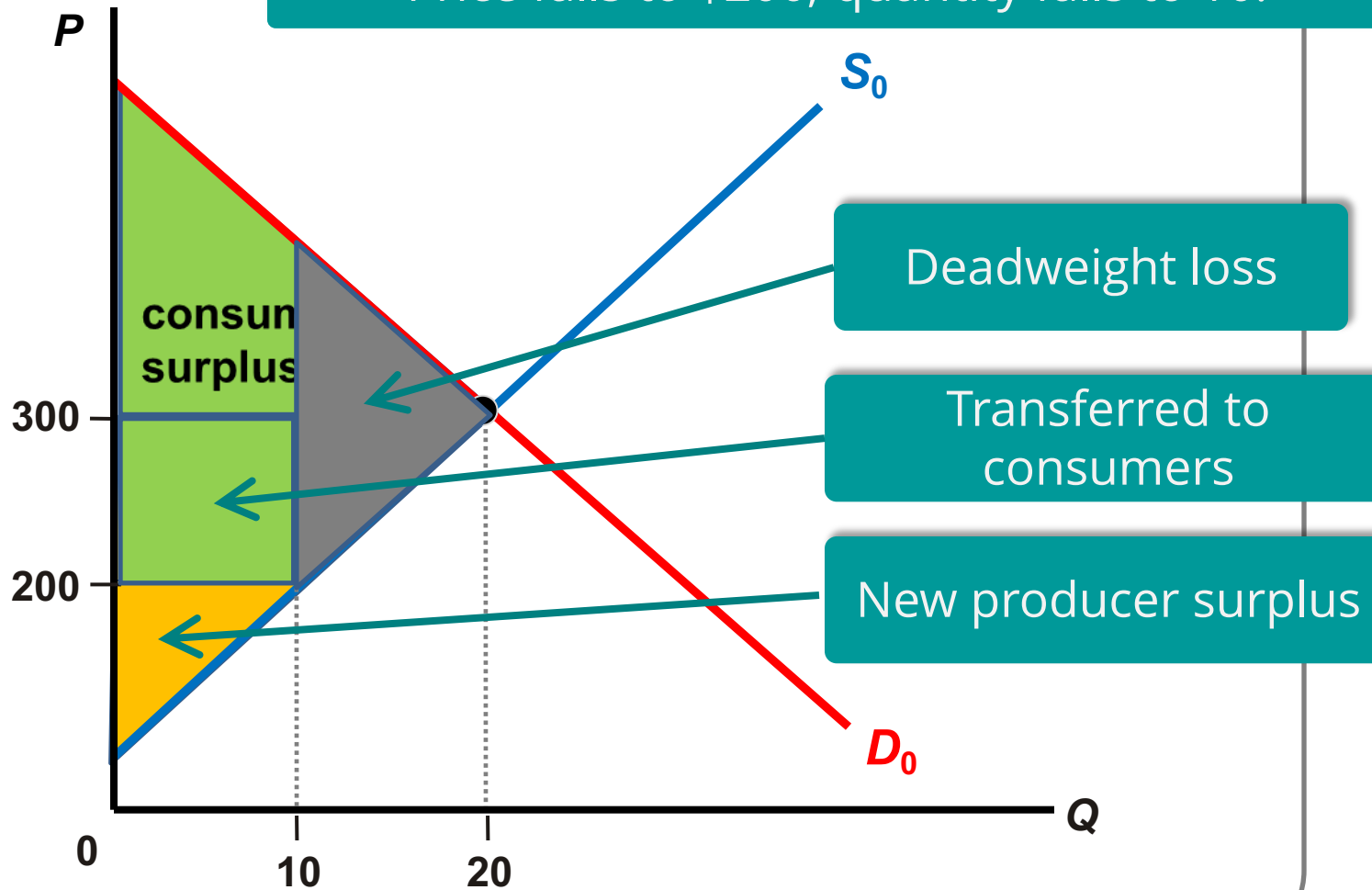
WHEN PRICES EXCEED EQUILIBRIUM

Price rises to \$400; quantity falls to 10.



WHEN PRICES ARE BELOW EQUILIBRIUM

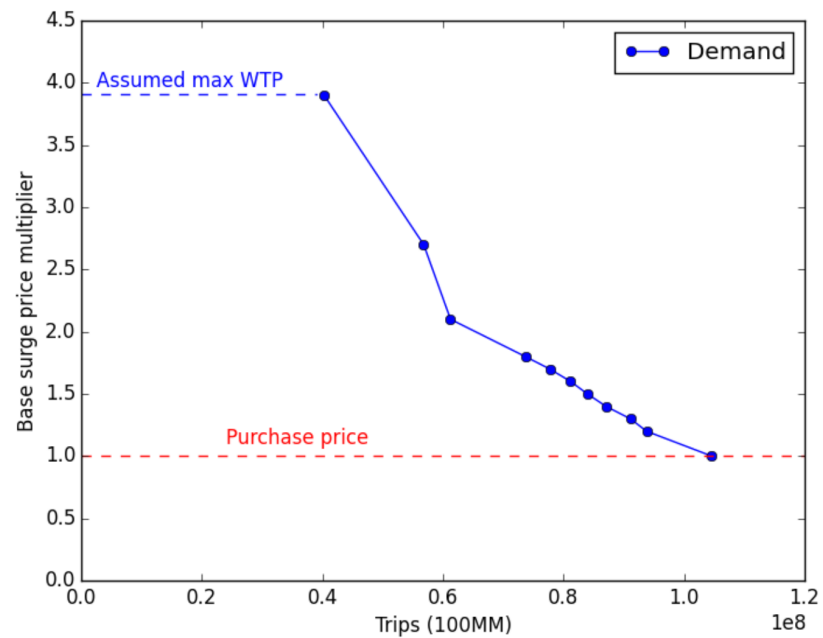
Price falls to \$200; quantity falls to 10.



America Would Happily Pay Uber An Extra \$7 Billion

Economists put a (big) number on the ride service's consumer surplus in 2015.

Figure 6: Visual representation of demand curve for transactions at 1.0x



Note: This figure presents a piecewise linear demand curve with jumps at each price discontinuity. The curve is generated from the underlying elasticities estimated for each price discontinuity and for consumers facing transactions at 1.0x.

Figure 5: Request rate drops at pricing discontinuities

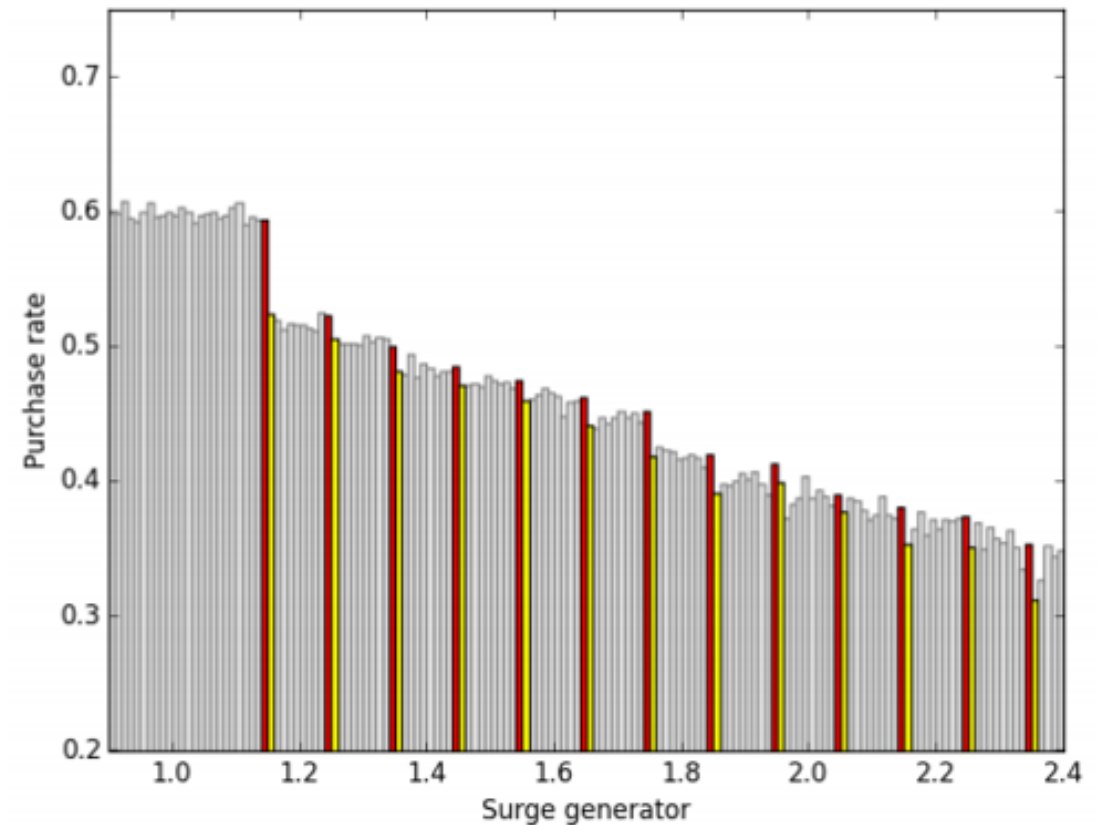


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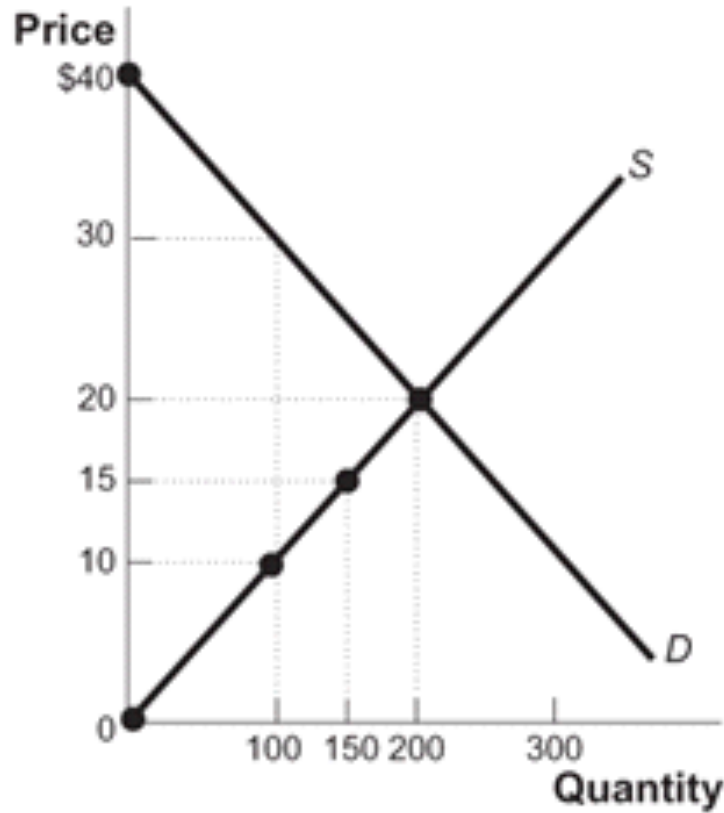
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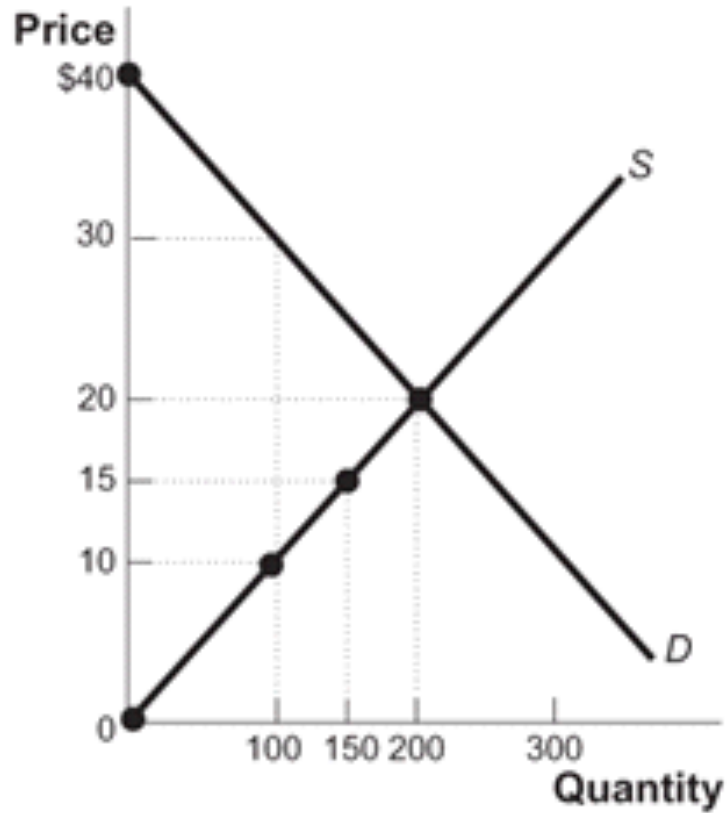
- a) \$50
- b) \$45
- c) \$150
- d) \$200



Look at the market for octopus candle holders. What is **producer surplus** at equilibrium?

- a) \$1000
- b) \$2000
- c) \$4000
- d) \$8000

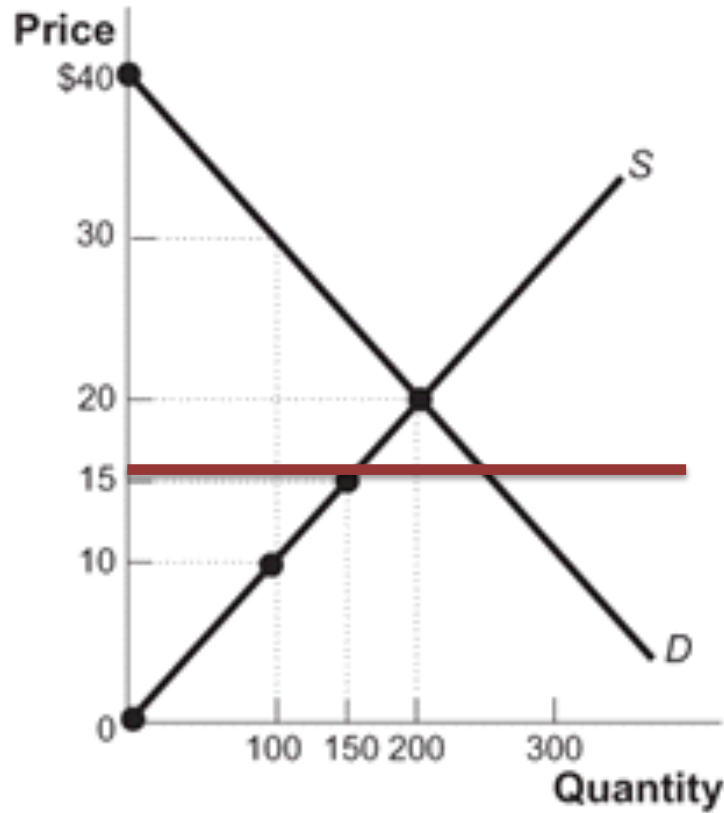




Look at the market for octopus candle holders. What is **total surplus** at equilibrium?

- a) \$1000
- b) \$2000
- c) \$4000
- d) \$8000





Amazon caps the price at \$15. What is the new **producer surplus**?

- a) \$1125
- b) \$1875
- c) \$2000
- d) \$2250



Efficiency of Markets

- Competitive markets are usually efficient:
 1. Allocate consumption to the potential **buyers who most value it.**
 2. Allocate sales to potential **sellers** who most value right to sell the good (e.g., **who have the lowest cost**).
 3. They ensure that **all transactions are mutually beneficial**

Why markets *typically* work so well

Well-functioning markets are effective because of:

- 1. Property rights:** the rights of owners of valuable items, whether resources goods, to use those items as they choose or.
- 2. Economic signals:** any piece of information that helps people make better economic decisions.

Four sources of market failures

Markets will not be efficient in the presence of **market failures!**

- Lack of competition
- Information not shared by all parties
- Existence of external benefits/costs (externalities)
- Existence of public goods